Comptia N+

N+ NETWORK FUNDAMENTALS AND BUILDING NETWORKS

DAY-3

TASKS:1

List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

RESULT

IPv4	IPv6
Uses 32-bit addresses.	Uses 128-bit addresses.
Address format: 192.168.1.1	Address format: 2001:db8::1
Supports about 4.3 billion addresses.	Supports a very large number of addresses (almost unlimited).
Uses decimal numbers separated by dots.	Uses hexadecimal numbers separated by colons.
Example: 192.168.10.25	Example: 2001:db8:85a3::8a2e:370:7334

TASKS:2

Given the following scenario: You have 4 devices (laptop, phone, smart

TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

RESULT

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

TASKS:3

Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

RESULT

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- Network: **192.168.1.0**
- CIDR: **/27** CALCULATION:
- Default subnet mask = **/24**
- New subnet mask = **/27**
- Borrowed bits = **3**
- Number of subnets = **$2^3 = 8$**
- Host bits = **5**
- Hosts per subnet = **$2^5 – 2 = 30$ usable hosts**

Subnet	Network Address	Usable Host Range	Broadcast Address
1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

TASKS:4

You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet. Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

RESULT

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- IP Address: **10.0.5.17**
- Subnet Mask: **255.255.255.0** CALVULATION:

Step 1: Convert to CIDR

255.255.255.0 = **/24**

Step 2: Network Address

10.0.5.0

Step 3: Broadcast Address

10.0.5.255

Step 4: Valid Host Range

First Host: 10.0.5.1

Last Host: 10.0.5.254

Item	**Value**
CIDR	/24
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
Valid Host Range	10.0.5.1 – 10.0.5.254

TASKS:5

Use an online subnet calculator tool (such as https://www.subnetcalculator.com/) to verify your answer for the

previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

RESULT

1. Opened the website https://www.subnet-calculator.com/ in a web browser.
2. Entered the IP address 10.0.5.17.
3. Entered the subnet mask 255.255.255.0.
4. Clicked Calculate.
5. Compared the results with my manual calculation.
6. Took a screenshot of the calculator output.

OBSERVATION:

The subnet calculator showed:

- Network Address: 10.0.5.0
- Broadcast Address: 10.0.5.255
- Host Range: 10.0.5.1 – 10.0.5.254

The result from the online subnet calculator matched manual calculation. This confirmed that my subnetting answer was correct."